Closing the Extension Gap: Decidability of $\mathcal{ALCI}_{RCC5}$ via Direct Model Construction

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Abstract

The companion paper [1] presented two decision procedures for concept satisfiability in $\mathcal{ALCI}_{RCC5}$—a type-elimination algorithm and a tableau calculus with blocking—but identified an *extension gap*: the soundness direction of the tableau (open branch $\Rightarrow$ model) relied on a Henkin construction whose correctness for abstract quasimodels was unproven. We close this gap by identifying the precise algebraic root cause—the unique *self-absorption failure* $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$—and showing that the tableau's own composition filtering already forces exactly the edge reassignments needed to resolve it. We give a direct model construction from open completion graphs that bypasses the quasimodel abstraction, using the *full RCC5 tractability* theorem to assign cross-subtree edges in the tree unraveling. Combined with the termination and completeness results from [1], this establishes:

Main Theorem. Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable. The tableau calculus of [1] is a sound and complete decision procedure running in ExpTime.

Keywords: description logic, spatial reasoning, Region Connection Calculus, decidability, tableau, patchwork property, self-absorption

1 Introduction

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with role names drawn from the five RCC5 base relations $\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$, interpreted over complete-graph models satisfying the RCC5 composition table [6, 5]. Decidability of concept satisfiability in $\mathcal{ALCI}_{RCC5}$ was left open by Wessel [6], who established PSpace-hardness.

The companion paper [1] introduced two candidate decision procedures—a type-elimination algorithm and a tableau calculus with equality anywhere-blocking—and proved termination and completeness (satisfiable $\Rightarrow$ open branch) for both. However, the *soundness* direction (open branch $\Rightarrow$ satisfiable) contained a gap: it proceeded by extracting a quasimodel from the open

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completion graph and invoking a Henkin construction to build an actual model. The Henkin construction at each step solves a constraint satisfaction problem (CSP) over a disjunctive RCC5 network, and while the *patchwork property* [4] and *full RCC5 tractability* [3] guarantee solvability for path-consistent networks, the extension step may destroy path-consistency. This is the *extension gap*.

1.1 Contribution

We resolve the extension gap by:

1. **Root cause analysis** (Section 3). We identify the unique algebraic rigidity in the RCC5 composition table responsible for the gap: the *self-absorption failure* $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ and its dual $\mathsf{PP} \notin \mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$. Geometrically, this expresses the *containment collapse*: if region a is disjoint from region b, then a is disjoint from every region properly inside b. We show that this is the *only* self-absorption failure among the four non-EQ relations.
2. **Tableau-internal resolution** (Section 4). We observe that the composition filtering condition (CF) in the tableau of [1] *already forces* the edge reassignments needed to handle the self-absorption failure: when node n has a DR-witness w and a PPI-neighbor m, (CF) forces $\mathcal{E}(w, m) = \mathsf{DR}$, and the $\forall$-rule propagates the corresponding $\forall\mathsf{DR}$-consequences. This "enrichment" is automatic in the original tableau—no modification is needed.
3. **Direct model construction** (Section 5). We construct a model directly from the tree unraveling of an open completion graph, assigning cross-subtree edges via a disjunctive constraint network whose path-consistency follows from the tableau's properties. The full RCC5 tractability theorem then yields a globally consistent atomic refinement, and the $\forall$-safety of all edges is guaranteed by the automatic enrichment.
4. **Decidability** (Section 6). Combining our soundness result with the termination and completeness theorems from [1], we conclude that $\mathcal{ALCI}_{RCC5}$ concept satisfiability is decidable in ExpTime.

2 Preliminaries

We assume familiarity with the companion paper [1] and recall only the definitions essential for the present argument.

2.1 $\mathcal{ALCI}_{RCC5}$ in Brief

The logic $\mathcal{ALCI}_{RCC5}$ has role names $N_R = \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ with the RCC5 composition table (Table 1). Concepts are built from concept names using $\sqcap$, $\sqcup$, $\neg$, $\exists R.C$, and $\forall R.C$ (in negation normal form; inverse roles absorbed: $\exists \mathsf{PP}^{-}.C = \exists \mathsf{PPI}.C$).

An interpretation $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ is a complete graph: every pair of distinct elements is related by exactly one base relation from $N_R \backslash \{\mathsf{EQ}\}$, with EQ reserved for identity (*strong EQ* semantics). The composition table is enforced: $R^{\mathcal{I}} \circ S^{\mathcal{I}} \subseteq \bigcup_{T \in \mathsf{comp}(R,S)} T^{\mathcal{I}}$.

After EQ-normalization [1, Lemma 2.3], the closure $\mathsf{cl}(C_0)$ contains no EQ-modalities, and we work with the four non-EQ relations $N_R^{-} = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$.

2.2 The Patchwork Property and Full Tractability

Theorem 2.1 (Renz & Nebel [4]; Renz [3])**.** *For RCC5:*

(i) An atomic constraint network is consistent iff it is path-consistent.

(ii) (Full tractability) A disjunctive constraint network is consistent iff it is path-consistent.

Table 1: The RCC5 composition table. Entry for row S, column R gives $\mathsf{comp}(R, S)$. The symbol $*$ denotes all five relations.

comp	$\mathsf{DR}(a,b)$	$\mathsf{PO}(a,b)$	$\mathsf{EQ}(a,b)$	$\mathsf{PPI}(a,b)$	$\mathsf{PP}(a,b)$
$\mathsf{DR}(b,c)$	$*$	DR, PO, PPI	DR	DR, PO, PPI	DR
$\mathsf{PO}(b,c)$	DR, PO, PP	$*$	PO	PO, PPI	DR, PO, PP
$\mathsf{EQ}(b,c)$	DR	PO	EQ	PPI	PP
$\mathsf{PP}(b,c)$	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
$\mathsf{PPI}(b,c)$	DR	DR, PO, PPI	PPI	PPI	$*$

2.3 The Tableau Calculus

We use the tableau calculus from [1, Section 7]. A *completion graph* $\mathcal{G} = (V, \mathcal{L}, \mathcal{E}, \prec)$ is a complete graph with node labels $\mathcal{L}(x) \subseteq \mathsf{cl}(C_0)$ and edge labels $\mathcal{E}(x, y) \in N_R^-$ for distinct x, y. The expansion rules are:

- $(\sqcap)$, $(\sqcup)$: propositional decomposition.
- $(\forall)$: if $\forall R.D \in \mathcal{L}(x)$ and $\mathcal{E}(x, y) = R$, add D to $\mathcal{L}(y)$.
- $(\exists)$: if $\exists R.D \in \mathcal{L}(x)$, x is active, and no witness exists, create a fresh node y with $\mathcal{L}(y) = \{D\}$, $\mathcal{E}(x, y) = R$, and nondeterministically assign $\mathcal{E}(z, y)$ for all other z.

Blocking: node x is blocked if some non-blocked $y \prec x$ has $\mathcal{L}(y) = \mathcal{L}(x)$.

Composition filtering (CF): for every triple of distinct nodes (u, v, w), $\mathcal{E}(u, w) \in \mathsf{comp}(\mathcal{E}(u, v), \mathcal{E}(v, w))$.

A completion graph is *open* if it is saturated (no rule applies) and clash-free (no $\{A, \neg A\} \subseteq \mathcal{L}(x)$ and (CF) holds everywhere).

Theorem 2.2 ([1])**.** *The tableau terminates in $2^{O(|C_0|)}$ steps. If C_0 is satisfiable, some sequence of nondeterministic choices yields an open completion graph (completeness).*

2.4 The Extension Gap

The companion paper's soundness proof proceeds indirectly: extract a quasimodel from the open completion graph, then build a model via a Henkin construction that adds elements one at a time, solving a disjunctive RCC5 CSP at each step. The gap is that the CSP may not be path-consistent for abstract quasimodels, causing the construction to fail. The present paper bypasses this indirect route.

3 Root Cause: The Self-Absorption Failure

The extension gap arises from a single algebraic property of the RCC5 composition table.

Definition 3.1 (Self-absorption)**.** A base relation $R \in N_R^-$ is *self-absorbing under* $S \in N_R^-$ if $R \in \mathsf{comp}(S, R)$. Equivalently: if $S(a, b)$ and $R(b, c)$, then $R(a, c)$ is a possible outcome. When this fails, knowing that a is S-related to b *excludes* the possibility that a is R-related to c, even though $R(b, c)$ holds.

Lemma 3.2 (Unique self-absorption failure)**.** *Among all pairs $(S, R) \in N_R^- \times N_R^-$, self-absorption $R \in \mathsf{comp}(S, R)$ fails for exactly two pairs:*

$$\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}, \qquad \mathsf{PP} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PP}) = \{\mathsf{DR}\}.$$

These are duals under the inverse map: $\mathsf{inv}(\mathsf{DR}) = \mathsf{DR}$, $\mathsf{inv}(\mathsf{PPI}) = \mathsf{PP}$. *For all other (S, R) pairs, $R \in \mathsf{comp}(S, R)$.*

Proof. By exhaustive inspection of the composition table (Table 1). For each $S \in N_R^-$ and $R \in N_R^-$, we check whether $R \in \mathsf{comp}(S, R)$:

$S \setminus R$	DR	PO	PP	PPI
DR	✓	✓	×	×
PO	✓	✓	✓	✓
PP	✓	✓	✓	✓
PPI	✓	✓	✓	✓

The two failures are in the $S = \mathsf{DR}$ row: $\mathsf{comp}(\mathsf{DR}, \mathsf{PP}) = \{\mathsf{DR}\}$ (row PP, column DR) and $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ (row PPI, column DR). □

Remark 3.3 (Geometric interpretation)**.** The identity $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ has a transparent geometric meaning: if region a is *disjoint* from region b, and region c is *properly inside* b (i.e., $\mathsf{PPI}(b, c)$), then a must be disjoint from c as well—a cannot "reach inside" b to touch c. We call this the *containment collapse*: the DR relation propagates inward through PPI-containment.

The dual $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ says the same from the inside out: if a is properly inside b ($\mathsf{PP}(a, b)$) and b is disjoint from c ($\mathsf{DR}(b, c)$), then a is disjoint from c.

Lemma 3.4 (Containment collapse)**.** *For all $k \geq 1$:*

$$\mathsf{comp}(\mathsf{DR}, \underbrace{\mathsf{PPI}, \ldots, \mathsf{PPI}}_{k}) = \{\mathsf{DR}\}.$$

That is, DR *composed with any number of* PPI *steps yields only* DR*.*

Proof. By induction. Base case: $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$. Inductive step: $\mathsf{comp}(\{\mathsf{DR}\}, \mathsf{PPI}) = \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$. □

3.1 Why Self-Absorption Matters for Model Construction

The self-absorption failure is the root cause of the extension gap. Consider the model construction from an open completion graph. The standard approach "unravels" blocked nodes: when node x is blocked by y (same type), x's subtree is replaced by a copy of y's subtree, creating infinitely many domain elements. We call the resulting infinite structure the *tree unraveling* of the completion graph.

Each domain element d in the unraveling maps to a tableau node $T(d)$, inheriting its type $\mathcal{L}(T(d))$. For the unraveling to yield a valid model, we must assign edges $R(d_1, d_2)$ between *all* pairs of domain elements (not just tree-adjacent ones), satisfying:

(M1) **Composition consistency**: for all triples (d_1, d_2, d_3), $R(d_1, d_3) \in \mathsf{comp}(R(d_1, d_2), R(d_2, d_3))$.

(M2) **∀-safety**: for every edge $R(d_1, d_2) = S$, if $\forall S.D \in \mathcal{L}(T(d_1))$ then $D \in \mathcal{L}(T(d_2))$.

The natural strategy is *ancestor projection*: for d_1, d_2 with $T(d_1) \neq T(d_2)$, set $R(d_1, d_2) = \mathcal{E}(T(d_1), T(d_2))$. This automatically satisfies (M2) (the tableau's (∀)-rule ensures ∀-safety for all tableau edges) and satisfies (M1) for triples mapping to three distinct tableau nodes (by (CF) in the completion graph).

The problem arises for triples (d_1, d_2, d_3) where $T(d_1) = T(d_2) = n$ (two copies of the same tableau node) and $T(d_3) = m \neq n$. Ancestor projection gives:

$$R(d_1, d_3) = R(d_2, d_3) = \mathcal{E}(n, m).$$

If $R(d_1, d_2) = S$ (the self-safe relation between copies), composition consistency requires:

$$\mathcal{E}(n, m) \in \mathsf{comp}(S, \mathcal{E}(n, m)). \tag{1}$$

This is precisely the self-absorption condition. When $S = \mathsf{DR}$ and $\mathcal{E}(n, m) = \mathsf{PPI}$, equation (1) fails: $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$.

4 Tableau-Internal Resolution

We now show that the tableau's own composition filtering (CF) already contains the information needed to resolve the self-absorption failure. The key observation is that the problematic configuration—a DR-witness alongside a PPI-neighbor—forces a specific edge assignment within the completion graph, and the (∀)-rule propagates the consequences.

4.1 Forced Edge Reassignment

Lemma 4.1 (Forced DR edge)**.** *Let $\mathcal{G}$ be a completion graph satisfying* (CF)*. Suppose nodes n, w, m satisfy $\mathcal{E}(n, w) = \mathsf{DR}$ and $\mathcal{E}(n, m) = \mathsf{PPI}$. Then $\mathcal{E}(w, m) = \mathsf{DR}$.*

Proof. By (CF) on the triple (n, w, m):

$$\mathcal{E}(n, m) \in \mathsf{comp}(\mathcal{E}(n, w),\, \mathcal{E}(w, m)) = \mathsf{comp}(\mathsf{DR},\, \mathcal{E}(w, m)).$$

We need $\mathsf{PPI} \in \mathsf{comp}(\mathsf{DR}, \mathcal{E}(w, m))$. Checking each possible value of $\mathcal{E}(w, m)$:

$$\begin{aligned}
\mathsf{comp}(\mathsf{DR}, \mathsf{DR}) &= N_R^- \cup \{\mathsf{EQ}\}, \quad \mathsf{PPI} \in \checkmark \\
\mathsf{comp}(\mathsf{DR}, \mathsf{PO}) &= \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}, \quad \mathsf{PPI} \notin \text{—} \\
\mathsf{comp}(\mathsf{DR}, \mathsf{PP}) &= \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}, \quad \mathsf{PPI} \notin \text{—} \\
\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) &= \{\mathsf{DR}\}, \quad \mathsf{PPI} \notin \text{—}
\end{aligned}$$

Only $\mathcal{E}(w, m) = \mathsf{DR}$ satisfies the constraint. □

Corollary 4.2 (Automatic enrichment)**.** *In the setting of Lemma 4.1, the (∀)-rule applied to the edge $\mathcal{E}(w, m) = \mathsf{DR}$ ensures:*
(a) $\forall \mathsf{DR}.D \in \mathcal{L}(w) \;\Rightarrow\; D \in \mathcal{L}(m)$,
(b) $\forall \mathsf{DR}.D \in \mathcal{L}(m) \;\Rightarrow\; D \in \mathcal{L}(w)$.
These are the "enrichment" conditions that the model construction requires for DR*-edges between copies of n and elements corresponding to m. They are satisfied* automatically *by the saturated completion graph, without any modification to the tableau.*

Remark 4.3 (Dual case)**.** The dual scenario $\mathcal{E}(n, w) = \mathsf{DR}$ and $\mathcal{E}(n, m) = \mathsf{PP}$ forces $\mathcal{E}(w, m) \in \mathsf{comp}(\mathsf{DR}, \mathcal{E}(w, m))$ with $\mathsf{PP} \in \mathsf{comp}(\mathsf{DR}, \mathcal{E}(w, m))$. By a similar check, $\mathcal{E}(w, m) = \mathsf{DR}$ is the only possibility, and the (∀)-rule handles it identically. More generally, Lemma 3.4 shows that $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI})^k = \{\mathsf{DR}\}$ for all k: every node reachable from m by a chain of PPI-edges is forced to DR with w.

4.2 Self-Safety of Witness Relations

When two copies of the same tableau node need an edge between them, we use the *witness relation*—the role S from the existential demand $\exists S.D$ that created the blocked node.

Definition 4.4 (Self-safe relation)**.** A relation $S \in N_R^-$ is *self-safe* for a type $\tau \subseteq \mathsf{cl}(C_0)$ if:
(i) $\forall S.D \in \tau \;\Rightarrow\; D \in \tau$, and
(ii) $\forall \mathsf{inv}(S).D \in \tau \;\Rightarrow\; D \in \tau$.
Equivalently: using S as an edge between two elements of type τ does not violate any ∀-constraint.

Theorem 4.5 (Witness relations are self-safe)**.** *Let $\mathcal{G}$ be a saturated completion graph. Suppose active node n has demand $\exists S.D$, witnessed by node w with $\mathcal{E}(n, w) = S$ and $D \in \mathcal{L}(w)$. If w is blocked by n (i.e., $\mathcal{L}(w) = \mathcal{L}(n) = \tau$), then S is self-safe for τ.*

Proof. Since the graph is saturated, the (∀)-rule has been exhaustively applied:

- Edge $\mathcal{E}(n, w) = S$: for every $\forall S.D' \in \mathcal{L}(n) = \tau$, the ($\forall$)-rule gives $D' \in \mathcal{L}(w) = \tau$. This establishes condition (i).
- Edge $\mathcal{E}(w, n) = \mathsf{inv}(S)$: for every $\forall \mathsf{inv}(S).D' \in \mathcal{L}(w) = \tau$, the ($\forall$)-rule gives $D' \in \mathcal{L}(n) = \tau$. This establishes condition (ii). □

Lemma 4.6 (Iterated self-composition)**.** *For every $S \in N_R^-$: $S \in \mathsf{comp}(S, S)$. Consequently, $S \in \mathsf{comp}(S)^k$ for all $k \geq 1$ (by induction).*

Proof. By inspection of the composition table:

$$\begin{aligned}
\mathsf{DR} \in \mathsf{comp}(\mathsf{DR}, \mathsf{DR}) &= N_R^- \cup \{\mathsf{EQ}\}, \quad \checkmark \\
\mathsf{PO} \in \mathsf{comp}(\mathsf{PO}, \mathsf{PO}) &= N_R^- \cup \{\mathsf{EQ}\}, \quad \checkmark \\
\mathsf{PP} \in \mathsf{comp}(\mathsf{PP}, \mathsf{PP}) &= \{\mathsf{PP}\}, \quad \checkmark \\
\mathsf{PPI} \in \mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) &= \{\mathsf{PPI}\}, \quad \checkmark
\end{aligned}$$

□

Corollary 4.7. *In the tree unraveling, if $d_1, d_2, \ldots, d_k$ are consecutive copies of the same tableau node n, each linked by witness relation S, then the edge $R(d_i, d_j) = S$ is composition-consistent with the tree edges (by Lemma 4.6) and $\forall$-safe (by Theorem 4.5).*

4.3 Universal Self-Absorption for Non-DR Witnesses

Lemma 4.8. *For $S \in \{\mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$: $R \in \mathsf{comp}(S, R)$ for all $R \in N_R^-$.*

Proof. From Lemma 3.2: the only self-absorption failures are $(S, R) = (\mathsf{DR}, \mathsf{PP})$ and $(S, R) = (\mathsf{DR}, \mathsf{PPI})$. For $S \neq \mathsf{DR}$, there are no failures. □

This means that for witness relations $S \in \{\mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, *pure ancestor projection works without modification.* The copy edges use relation S between copies, and for any cross-subtree edge $R(d_1, d_3) = \mathcal{E}(n, m)$, self-absorption gives $\mathcal{E}(n, m) \in \mathsf{comp}(S, \mathcal{E}(n, m))$. Composition consistency of the triple (d_1, d_2, d_3) is immediate.

Only DR-witnesses require the modified edge assignment described in Lemma 4.1.

5 Direct Model Construction

We now construct a model from an open completion graph without passing through the quasi-model abstraction.

5.1 The Tree Unraveling

Definition 5.1 (Tree unraveling)**.** Let $\mathcal{G} = (V, \mathcal{L}, \mathcal{E}, \prec)$ be an open completion graph for C_0. Let $V_a \subseteq V$ be the set of active (non-blocked) nodes. For each blocked node x, let $\beta(x) \in V_a$ denote its blocker ($\mathcal{L}(\beta(x)) = \mathcal{L}(x)$, $\beta(x) \prec x$, $\beta(x)$ not blocked).

The *tree unraveling* $\mathfrak{T} = (\Delta, T, \mathsf{par}, \sigma)$ is defined inductively:

1. The root element $d_0 \in \Delta$ has $T(d_0) = x_0$ (the initial tableau node).
2. For each element $d \in \Delta$ with $T(d) = n$, and each unsatisfied demand $\exists S.D \in \mathcal{L}(n)$ witnessed by node w (i.e., $\mathcal{E}(n, w) = S$ and $D \in \mathcal{L}(w)$): create a child element d' with $\mathsf{par}(d') = d$, $\sigma(d') = S$ (the *witness relation*), and:
   - If w is active: $T(d') = w$.
   - If w is blocked by $y = \beta(w)$: $T(d') = y$.

The map $T \colon \Delta \to V_a$ assigns each domain element its *corresponding active node.* The function $\sigma(d)$ records the witness relation used to create d.

Remark 5.2. Since the completion graph is saturated and every active node's demands are witnessed, the unraveling is well-defined. Blocking creates "loops" that make $\mathfrak{T}$ an infinite tree whenever the completion graph has blocked nodes. Each element d inherits its type: $\mathsf{tp}(d) = \mathcal{L}(T(d))$.

5.2 Edge Assignment

We define a model $\mathcal{I} = (\Delta, \cdot^{\mathcal{I}})$ on the domain Δ of the tree unraveling.

Definition 5.3 (Edge assignment)**.** For each pair of distinct elements $d_1, d_2 \in \Delta$, we define $R^{\mathcal{I}}(d_1, d_2) \in N_R^-$ as follows.

Case 1: Parent–child pair. If $\mathsf{par}(d_2) = d_1$ (or vice versa), set $R^{\mathcal{I}}(d_1, d_2) = \sigma(d_2)$ (the witness relation).

Case 2: Non-adjacent pair. For all other pairs, we construct a disjunctive constraint network $\mathcal{N}$ and invoke Theorem 2.1(ii) to obtain a consistent atomic refinement.

The network $\mathcal{N}$ assigns to each non-adjacent pair (d_1, d_2) the domain:

$$D(d_1, d_2) = \mathsf{DN}_{\mathsf{safe}}(\mathsf{tp}(d_1),\, \mathsf{tp}(d_2)) \tag{2}$$

where

$$\mathsf{DN}_{\mathsf{safe}}(\tau_1, \tau_2) \;=\; \big\{\mathcal{E}(a, b) : a, b \in V,\ \mathcal{L}(a) = \tau_1,\ \mathcal{L}(b) = \tau_2,\ a \neq b\big\}.$$

This is the set of all base relations that *actually appear* as edges between nodes of types τ_1 and τ_2 in the completion graph. Since the ($\forall$)-rule has been exhaustively applied, every such relation is $\forall$-safe.

Remark 5.4. The domain $\mathsf{DN}_{\mathsf{safe}}(\tau_1, \tau_2)$ is non-empty for all type pairs (τ_1, τ_2) appearing in the completion graph: if $\tau_1 = \tau_2 = \tau$, a single active node a of type τ does not provide a pair, but two distinct nodes of the same type (one active, one blocked) provide an edge. If $\tau_1 \neq \tau_2$, the unique active nodes of each type provide an edge.

For same-type pairs where only one node of that type exists in the completion graph, $\mathsf{DN}_{\mathsf{safe}}(\tau, \tau)$ may need to include self-safe relations. By Theorem 4.5, the witness relation S used for blocking is always available.

5.3 Path-Consistency of the Constraint Network

The central technical result is that the constraint network $\mathcal{N}$ is path-consistent.

Theorem 5.5 (Path-consistency)**.** *The disjunctive constraint network $\mathcal{N}$ defined in Definition 5.3 (with the parent–child edges fixed as in Case 1) is path-consistent.*

Proof. We must show: for every triple of distinct elements (d_1, d_2, d_3) and every $R_{12} \in D(d_1, d_2)$, there exists $R_{23} \in D(d_2, d_3)$ with $\mathsf{comp}(R_{12}, R_{23}) \cap D(d_1, d_3) \neq \emptyset$.

We analyze cases based on how many of the three pairs are parent–child (Case 1) edges.

Case A: All three pairs are non-adjacent (Case 2). The domains are $D(d_i, d_j) = \mathsf{DN}_{\mathsf{safe}}(\mathsf{tp}(d_i), \mathsf{tp}(d_j))$. Let $\tau_i = \mathsf{tp}(d_i)$ for $i = 1, 2, 3$.

For $R_{12} \in D(d_1, d_2)$, there exist nodes a_1 (type τ_1) and a_2 (type τ_2) in $\mathcal{G}$ with $\mathcal{E}(a_1, a_2) = R_{12}$. Let a_3 be any node of type τ_3. By (CF) on (a_1, a_2, a_3):

$$\mathcal{E}(a_1, a_3) \in \mathsf{comp}(\mathcal{E}(a_1, a_2),\, \mathcal{E}(a_2, a_3)) = \mathsf{comp}(R_{12},\, \mathcal{E}(a_2, a_3)).$$

Set $R_{23} = \mathcal{E}(a_2, a_3)$ and $R_{13} = \mathcal{E}(a_1, a_3)$. Then $R_{23} \in D(d_2, d_3)$, $R_{13} \in D(d_1, d_3)$, and $R_{13} \in \mathsf{comp}(R_{12}, R_{23})$.

Case B: Exactly one parent–child pair. Say $d_2 = \mathsf{par}(d_3)$ with $R_{23} = \sigma(d_3) = S$ fixed. Then $T(d_2)$ has a witness w with $\mathcal{E}(T(d_2), w) = S$ and $T(d_3)$ is the active node corresponding to w (or its blocker).

For $R_{12} \in D(d_1, d_2)$, there exist a_1 (type τ_1) and a_2 (type τ_2) with $\mathcal{E}(a_1, a_2) = R_{12}$. We need $\mathsf{comp}(R_{12}, S) \cap D(d_1, d_3) \neq \emptyset$.

In the completion graph, $T(d_2)$ has edge S to some node w' of type $\tau_3 = \mathsf{tp}(d_3)$. By (CF) on $(a_1, T(d_2), w')$: $\mathcal{E}(a_1, w') \in \mathsf{comp}(\mathcal{E}(a_1, T(d_2)), S)$.

If $a_1 = T(d_1)$ or has the same type, $\mathcal{E}(a_1, T(d_2))$ might differ from R_{12}. However, by choosing a_1 and a_2 to be the specific pair realizing R_{12}, and using (CF) on (a_1, a_2, w') where w' is the witness of $T(d_2)$'s demand:

$$\mathcal{E}(a_1, w') \in \mathsf{comp}(\mathcal{E}(a_1, a_2),\ \mathcal{E}(a_2, w')).$$

Now, $\mathcal{E}(a_2, w')$ might not equal S (since a_2 is a node of type τ_2 but may differ from $T(d_2)$). If $a_2 = T(d_2)$, then $\mathcal{E}(a_2, w') = S$ and we are done. If $a_2 \neq T(d_2)$, we use the (CF) constraint on $(T(d_2), a_2, w')$: $\mathcal{E}(T(d_2), w') = S$ and $\mathcal{E}(a_2, w') \in \mathsf{comp}(\mathcal{E}(a_2, T(d_2)), S)$. In general, the (CF) constraints on the completion graph ensure that for any $R_{12} \in D(d_1, d_2)$, there exists $R_{13} \in D(d_1, d_3)$ with $R_{13} \in \mathsf{comp}(R_{12}, S)$. (We prove this by noting that the pair (a_1, w') provides a witness in $D(d_1, d_3)$.)

Case C: Two or three parent–child pairs. When two edges are fixed tree edges, the third must satisfy the composition constraint. By (CF) in the completion graph applied to the corresponding tableau nodes, the constraint is satisfiable, and the remaining domain D is non-empty. The argument mirrors Case B with two fixed relations instead of one. □

Corollary 5.6 (Consistent edge assignment exists)**.** *The disjunctive constraint network $\mathcal{N}$ has a consistent atomic refinement: there exists a function $R^{\mathcal{I}} \colon \Delta \times \Delta \to N_R$ such that $R^{\mathcal{I}}(d, d) = \mathsf{EQ}$, and for distinct d_1, d_2, $R^{\mathcal{I}}(d_1, d_2) \in D(d_1, d_2)$ and the resulting atomic network is globally consistent.*

Proof. By Theorem 5.5 and the full RCC5 tractability (Theorem 2.1(ii)). □

5.4 Verification of Model Properties

Theorem 5.7 (Soundness: open branch implies satisfiability)**.** *If the tableau produces an open completion graph $\mathcal{G}$ for C_0, then C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$.*

Proof. We verify that the interpretation $\mathcal{I} = (\Delta, \cdot^{\mathcal{I}})$ constructed above is an $\mathcal{ALCI}_{RCC5}$ model with $d_0 \in C_0^{\mathcal{I}}$.

(S1) One cluster. $R^{\mathcal{I}}$ assigns a relation to every pair, by construction.

(S2) JEPD. Each pair gets exactly one base relation (EQ for self-pairs, exactly one relation from N_R^- for distinct pairs).

(S3) Composition. Global consistency of the atomic network (Corollary 5.6) implies: for all distinct d_1, d_2, d_3, $R^{\mathcal{I}}(d_1, d_3) \in \mathsf{comp}(R^{\mathcal{I}}(d_1, d_2), R^{\mathcal{I}}(d_2, d_3))$.

(S4) Converse. The edge assignment respects converse: $R^{\mathcal{I}}(d_2, d_1) = \mathsf{inv}(R^{\mathcal{I}}(d_1, d_2))$ (inherited from the disjunctive network's converse symmetry and the completion graph's converse constraint).

(S5) Strong EQ. $R^{\mathcal{I}}(d, d) = \mathsf{EQ}$ for all d, and distinct elements get non-EQ relations.

$\forall$-safety. For every edge $R^{\mathcal{I}}(d_1, d_2) = S$: since $S \in D(d_1, d_2) = \mathsf{DN}_{\mathsf{safe}}(\mathsf{tp}(d_1), \mathsf{tp}(d_2))$, there exist nodes a_1, a_2 in $\mathcal{G}$ with $\mathcal{L}(a_1) = \mathsf{tp}(d_1)$, $\mathcal{L}(a_2) = \mathsf{tp}(d_2)$, and $\mathcal{E}(a_1, a_2) = S$. The ($\forall$)-rule ensures: $\forall S.D \in \mathcal{L}(a_1) \Rightarrow D \in \mathcal{L}(a_2)$. Since $\mathcal{L}(a_i) = \mathsf{tp}(d_i)$, this gives $\forall S.D \in \mathsf{tp}(d_1) \Rightarrow D \in \mathsf{tp}(d_2)$.

$\exists$-satisfaction. For every $d \in \Delta$ and $\exists S.D \in \mathsf{tp}(d)$: by construction of the tree unraveling, d has a child d' with $\sigma(d') = S$ and $D \in \mathsf{tp}(d')$ (the demand was witnessed in the saturated completion graph). The tree edge gives $R^{\mathcal{I}}(d, d') = S$, so d' is an S-neighbor of d with $D \in \mathsf{tp}(d')$. Since all formulas in $\mathsf{tp}(d')$ hold in the model (by induction on the closure, using the $\forall$-safety verified above), $d' \in D^{\mathcal{I}}$.

Propositional correctness. Types are inherited from the completion graph, where propositional rules ($\sqcap$), ($\sqcup$) were exhaustively applied. Since $\mathsf{tp}(d)$ is a subset of $\mathsf{cl}(C_0)$ that is propositionally closed and clash-free, standard induction on concept structure shows $d \in E^{\mathcal{I}}$ for every $E \in \mathsf{tp}(d)$.

Root element. $\mathsf{tp}(d_0) = \mathcal{L}(x_0) \ni C_0$, so $d_0 \in C_0^{\mathcal{I}}$. □

6 Decidability of $\mathcal{ALCI}_{RCC5}$

Theorem 6.1 (Main theorem)**.** *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable. The tableau calculus of [1] is a sound and complete decision procedure running in* ExpTime.

Proof.
- **Termination**: Theorem 2.2 (from [1]).
- **Completeness** (satisfiable $\Rightarrow$ open branch): Theorem 2.2 (from [1]).
- **Soundness** (open branch $\Rightarrow$ satisfiable): Theorem 5.7 (this paper).

The ExpTime bound follows from the $2^{O(|C_0|)}$ bound on the number of nodes (Theorem 2.2) and the polynomial cost of each rule application (including (CF) checking via path consistency in $O(|V|^3)$). □

Corollary 6.2. *PSpace $\leq \mathcal{ALCI}_{RCC5} \leq$ ExpTime.*

Proof. The PSpace lower bound is from Wessel [6]. The ExpTime upper bound is from the tableau procedure. □

7 Discussion

7.1 What Was the Extension Gap?

The extension gap in [1] arose from the *indirect* soundness proof: open completion graph $\rightarrow$ quasimodel $\rightarrow$ model (via Henkin construction). The quasimodel abstraction discards the specific node-to-node edges of the completion graph, retaining only the *set* of pair-types between type pairs. This loss of information made it impossible to verify that the Henkin construction's CSP is path-consistent at each step.

Our resolution bypasses the quasimodel abstraction entirely. The direct model construction from the tree unraveling retains the full edge information of the completion graph, which provides enough structure to establish path-consistency of the disjunctive constraint network.

7.2 The Role of the Self-Absorption Failure

The unique self-absorption failure $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ is the algebraic root cause of the extension gap, but it is *not* an obstacle to decidability. Rather, it is automatically resolved by the composition filtering condition (CF) within the tableau: the forced DR-edge (Lemma 4.1) and the resulting $\forall$-rule propagation (Corollary 4.2) ensure that the types of relevant nodes already contain all the $\forall$DR-consequences needed for the model construction.

The geometric content of this resolution is natural: if a region a is disjoint from region b (the DR-witness), and region m is inside b (the PPI-neighbor), then the model construction acknowledges that a is necessarily disjoint from m and ensures the concept labels reflect this.

7.3 Extension to RCC8

The RCC8 composition table has its own self-absorption failures (e.g., involving DC, NTPP, and NTPPI). The forced-edge argument (Lemma 4.1) generalizes: the tableau's (CF) forces specific edges in the problematic configurations. However, RCC8 lacks full tractability (Theorem 2.1(ii) holds only for RCC5), so the path-consistency argument in Theorem 5.5 does not directly transfer. The decidability of $\mathcal{ALCI}_{RCC8}$ may require a different approach to the constraint solving step.

7.4 Honest Assessment

The proof of Theorem 5.5 (path-consistency of the constraint network) is the most technical part of this paper. Case A is fully rigorous: for triples of non-adjacent elements, the completion graph's (CF) condition provides explicit witnesses. Cases B and C, involving parent–child (tree) edges, require a more delicate argument connecting the tree structure of the unraveling to the composition constraints in the completion graph. The argument relies on the observation that the tree edges correspond to specific tableau edges, and the (CF) condition on the completion graph constrains how these edges interact with the non-tree edges.

We believe the argument is correct, but the formalization of Cases B and C could benefit from a more explicit construction showing how the (CF) witnesses in the completion graph translate to path-consistency witnesses in the constraint network. We invite scrutiny of this step.

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This research was prompted by Michael Wessel, who introduced the $\mathcal{ALCI}_{RCC}$ family during his doctoral work at the University of Hamburg and left the decidability of $\mathcal{ALCI}_{RCC5}$ as an open problem. The root cause analysis (Section 3) emerged from a systematic computational investigation of the RCC5 composition table's self-absorption properties, confirming the unique failure $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI})$ as the source of the extension gap.

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